PS200C: Quantitative Methods

In-class Activity 7: Difference-in-Differences

Name: ____________________

Question 1. Which of the following "modern DID" approaches can rescue your analysis if the **parallel trends** assumption is violated? (Circle all that apply.)

(a) Sun & Abraham (2021) cohort-specific event study

(b) Callaway & Sant'Anna (2021) group-time $ATT(g, t)$

(c) Borusyak, Jaravel & Spiess (2021) / Liu, Wang & Xu (2022) imputation (`fect`)

(d) de Chaisemartin & D'Haultfœuille (2020) convex-weighted clean 2×2s

(e) None of the above

Setup for Q2–Q4. In January 2018, Seattle imposed a sugary-drink tax. Portland did not. You have annual per-capita sales of sugary drinks (liters per person per year) in both cities, averaged over a short pre-window (2016–17) and a short post-window (2018–19):

	Pre-tax (2016–17)	**Post-tax (2018–19)**
Seattle	100 liters	85 liters
Portland	95 liters	88 liters

Question 2: Compute the DID. Using the four means above, compute the difference-in-differences estimate of the effect of the tax on Seattle's per-capita sugary-drink sales. Show your work.

Question 3: Where might parallel trends fail here?

(a) Explain what parallel trends requires us to believe here.

(b) Describe one concrete reason why parallel trends might fail. *Hint: think about something that could be changing during 2016–2019 that would influence both the sugary-drink-sales trend (absent the tax) and whether Seattle adopted the tax.*

Question 4: Pre-trends check. Your collaborator plots five years of pre-tax data (2013–2017) and shows that Seattle and Portland's per-capita sales were declining at nearly the same rate over that window. They conclude, "Parallel trends is satisfied—we're good." Is that right? Why or why not?